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Shibbolising UK Census and ESDS services

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Today

- Our project: SAFARI
- Background: the one-stop registration service
- Introducing Shibboleth
- A model of target-to-target communication
- Comparisons
- What next?

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SAFARI UKDA

- **S**hibboleth **A**uthentication **F**or **A**ccess to the **R**esource **I**nfrastructures of the **UKDA**
- JISC-funded, 1-year project (April 2005 – March 2006)
- Aims:
  - to apply Shibboleth middleware to 3 UKDA resources
  - to embed Shibboleth technology within the one-stop registration system which is used by MIMAS, EDINA, AHDS History – all geographically dispersed
    - this should provide greater flexibility for the resource owners and, consequently, for the users

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One-stop registration

- Established 2001/2002
  - Single registration for 7 resources
- Centralised registration but distributed logins
- Uses centralised Athens authentication / authorisation
- Communication occurs between resources, using strings written to users' Athens 'profiles' to identify:
  - Registered users
  - Users who have agreed to certain special conditions
  - Big disadvantage: this information cannot be over-written

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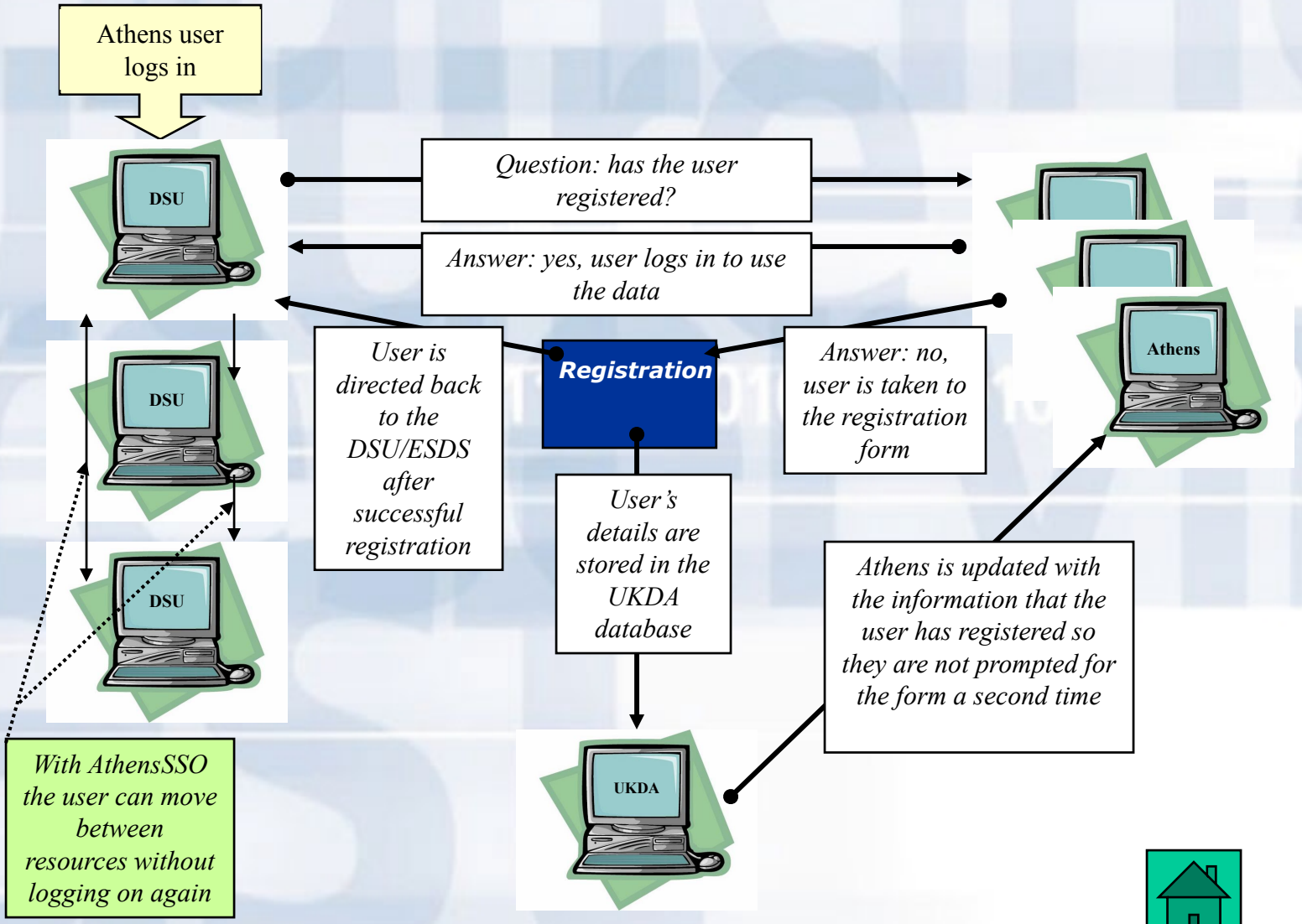
The current system



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The key issue

- How to enable the dispersed resources to recognise registered users using Shibboleth (and in a way that can be over-written)?

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Introducing Shibboleth

- Middleware which provides protocols for transferring user attributes from origin to target
- User authenticates locally
  - Method used is up to the origin
- User is authorised at the target
  - Method used is up to the resource, but using the attributes supplied by the origin – these attributes will be the key for us
  - Tools are available to help, such as PERMIS

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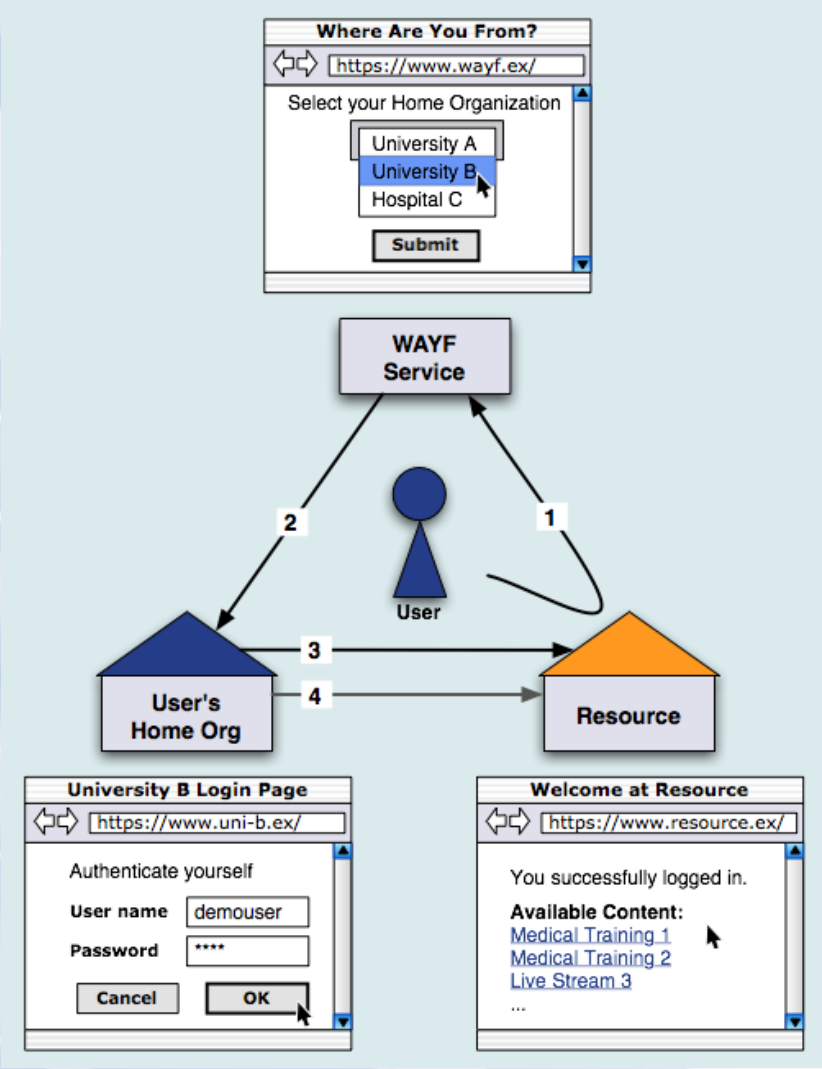
Shibboleth login



Picture courtesy of SWITCH, Swiss Education & Research Network, http://www.switch.ch/aai/demo/demo_intro.html

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Issues relating to Shibboleth

- Centralised registration = "making a cat bark"
  - Shibboleth emphasises user privacy
  - However, data owners require more controls
  - Others have approached similar resource-related identity issues
- Other advantages:
  - More fine-grained authorisation
  - Bring access control back to the resource owner
  - Complementary to Athens system, an alternative route into the data

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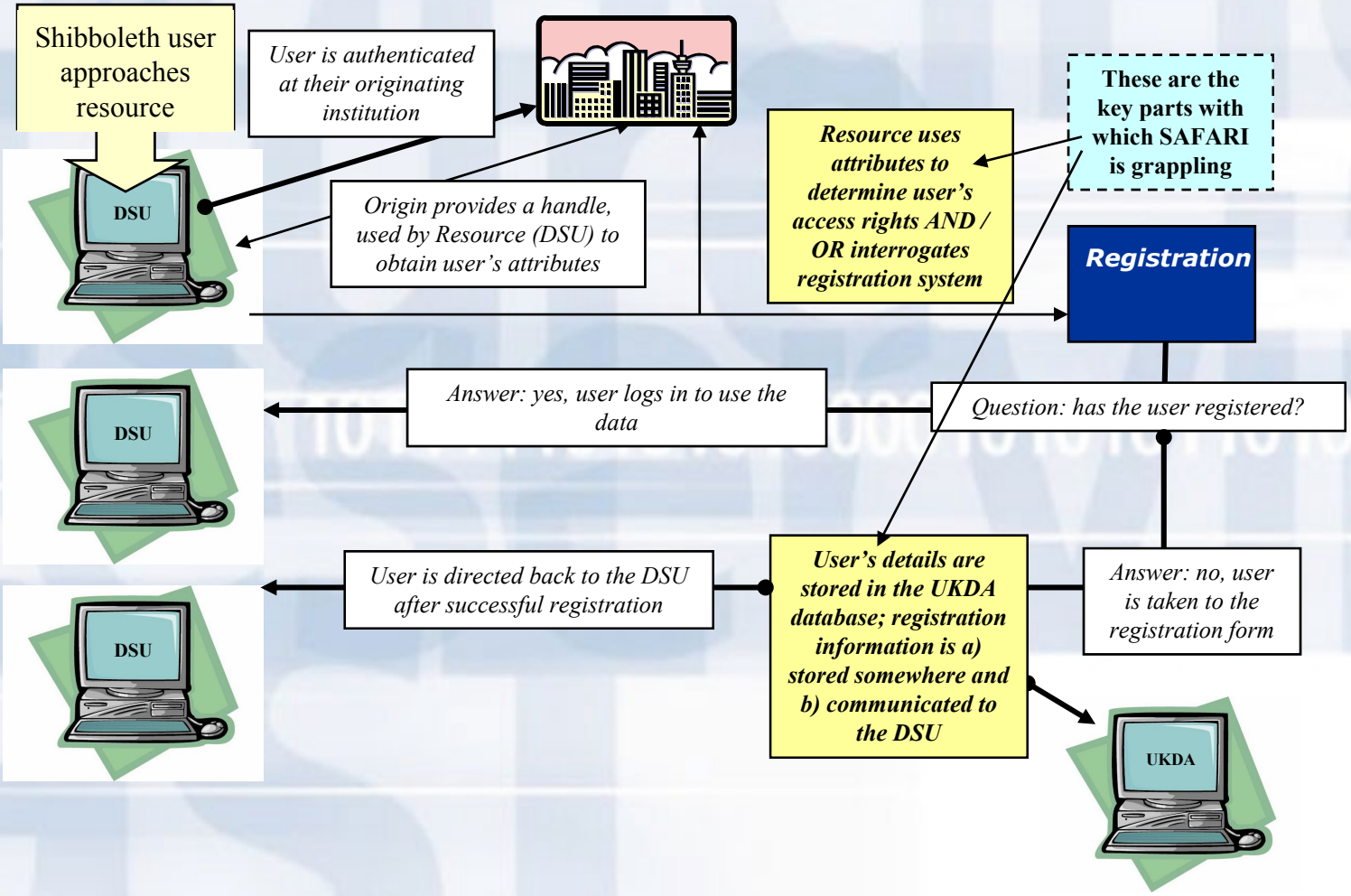
The potential system



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Target-to-target communication

- Possible solutions:
  - A separate call to Essex
    - Would ensure control was within the system's hands
    - But not necessarily standards-based
    - Doubles the risk of failure
    - Could employ session variables to allow single-sign on within one browser session
  - Establish a proxy Identity Provider (IdP)
    - Attractive alternative, but untested
    - Apply additional attributes for each resource, when logged into
    - Standards-based and recommended by Scott Cantor
    - Would still require an extra call to Essex at each authentication, which doubles the risk of failure

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Comparisons already in place

- VOSP (University of Alabama)
  - Similar to the idea of a proxy IdP
  - VO = Service Provider within a federation
  - User is directed to their local IdP ***and*** the VOSP to gather up all the attributes required before being authorised by the resource
  - Central attribute repository service for the VO
- SWITCHaai, Switzerland
  - Registration devolved to each organisation (7 signed up so far)
    - 'Registered User' = an attribute which origin supplies
  - Ideal but currently impractical in the UK (700+ organisations, as opposed to 7)
  - Problems of updating a registration module plug-in: special conditions would need to be stable etc.
- Shibboleth also in production standard via HAKA Federation, Finland

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What next?

- Finalise and implement the system specification
  - Complete the establishment of the 3 targets (underway)
  - Introduce the system of target-to-target communication (Sept 05)
  - Embed within the one-stop registration system (Dec 05)
- Evaluate, via user testing/survey (Jan/Feb 06), and refine
- Write up as a case study (Feb/Mar 06)
- If model created successfully, roll out to the remaining services (post Mar 06)

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More information

- Project Manager: Lucy Bell (lajbell@essex.ac.uk)
- Project web site: http://safari.data-archive.ac.uk (available June 05)

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